{COLORIO}

CPU
PI0
PI1
COLORCS
MREAD1
IOCS
MRW
HFLIP
FIXCS
MREAD2
MLWR
MUWR
EXRES
A[1..20]
D[0..15]
DMAINIT
VFLIP
OCHAROMCS
OBJSCRH
OBJSCRV
OBJCS
TIM1
VBLANK
CPUCLK
CPURES
FIXDTAC
MINT16EN
DMACS
V1SCRH
V2SCRH
SVCHAROMCS
VRCSET
SCVRAMCS
SYSWR
MULWR
File: cpu.kicad_sch

Timing
DMACS
FIXCS
HFLIP
VFLIP
RST
P4H
{HVCOUNTERS}
X1S
VCNTEN
D0
CLK2
CLK21
CLK22
CLR
P1H
P2H
N2H
N2HP4H
8H
P256VD
TIM
TIM3
OBJBUFINIT
DMAINIT
DMACLR
OBJBUFCLR
OSCANCLR
FBREF
CSYNC
CBLK
CS1
CS2
RS1
RS2
OE1
OE2
HSYNC
WR
EXRES
VBLANK
CPUCLK
CPURES
FIXDTAC
1VD
SYNCRLDV
TIM1
DMA
RAS
File: timing.kicad_sch

Sprites1
File: sprites1.kicad_sch
CLK21
CLR
P1H
P2H
N2H
N2HP4H
8H
P256VD
TIM
TIM3
OBJBUFINIT
DMAINIT
DMACLR
OBJBUFCLR
OSCANCLR
FBREF
HFLIP
A[1..19]
D[0..15]
MLWR
MUWR
MREAD2
OCHAROMCS
OBJSCRH
OBJSCRV
OBJCS
DMAON
CRTKILL
DMAD

GFX – Fixed layer
CLK2
P1H
P2H
P4H
{HVCOUNTERS}
TIM3
HFLIP
FIXCS
MREAD2
MLWR
MUWR
A[1..11]
D[0..15]
FX[0..7]
File: GFX–Fixed layer.kicad_sch

Sprites 2
CLK21
CLK22
CLR
P2H
N2H
P256VD
X1S
VCNTEN
OCHF
FBWM1
FBWM2
VD[0..7]
HD[2..8]
HJ[0..1]
W[0..15]
VPD8
C[0..3]
CX[0..3]
HFLIP
VFLIP
CS1
CS2
RS1
RS2
OE1
OE2
HSYNC
WR
File: sprites2.kicad_sch

GFX – Tiles
CLK2
EXRES
CLR
P2H
P4H
N2H
P1H
N2HP4H
1V
VCNTEN
SYNCRLDV
TIM1
RAS
HFLIP
HSYNC
V1SCRH
V2SCRH
SVCHAROMCS
VRCSET
SCVRAMCS
SYSWR
MLWR
MUWR
MULWR
MREAD2
A[1..17]
D[0..15]
CRTKILL
1C[0..6]
2C[0..6]
File: GFX–Tiles.kicad_sch

Color mixer
CLK2
A[1..12]
D[0..7]
MLWR
FX[0..7]
OCD[0..7]
1C[0..6]
2C[0..6]
PI0
PI1
COLORCS
MREAD1
CBLK
CRTKILL
RED
GREEN
BLUE
File: color_mixer.kicad_sch

IO
SPEAKER+
SPEAKER–
BSY
BUSY
RED
GREEN
BLUE
A[1..16]
D[0..7]
IOCS
EXRES
MRW
CSYNC
RST
MINT16EN
DMAON
CRTKILL
SOUND–ON
SOUND–DATA
SD0
File: io.kicad_sch

Sound
EXRES
SOUND–ON
SOUND–DATA
SD0
D[0..7]
SPEAKER+
SPEAKER–
BSY
BUSY
File: sound.kicad_sch

Timing diagrams
File: timing_diagrams.kicad_sch

Sprite timing diagrams
File: sprite_timing_diagrams.kicad_sch

U216D
74LS04
M11

U228B
74LS21
E19

U205E
74LS07
H29

U197C
74LS07
A22

U205D
74LS07
H29

U205F
74LS07
H29

U226C
74LS08
J27

U226B
74LS08
J27

A23F
74LS07

U198A
74LS32
F15

U225B
74LS08
D30

U225A
74LS08
D30

U210A
74LS74
J23

Ulf Skutnabba, twitter: @skutis77
Konami GX870
Sheet: /
File: fround.kicad_sch
Title: The Final Round

Size: A3	Date: 2024–09–08	**Rev:**
KiCad E.D.A. 8.0.5		Id: 1/12

RA2
4.7k x 4
G31
MHALT
MRESET
VCC

RA5
4.7k x 8
E31
MLDS
MUDS
MAS
MRW
MDTACK
MBGACK
MBR
MBERR
VCC

U229
74LS367
H30
CPUCLK
3-StateHexBuffers
GND

G31
R48
1k
VCC

U222A
74LS74
F20
MINT16END
VBLANKD
GND
VCC

U122
68000D
E30
CLK
IPL0
IPL1
IPL2
MBGACK
BGACK
MBR
BG
BR
FC0
FC1
FC2
VMA
E
VPA
BERR
MBERR
DTACK
MDTACK
HALT
MHALT
RESET
RMC
AS
UDS
LDS
R/W
DTACK
A1..A23
D0..D15
MA1..MA14
A15..A20
MA21..MA23
MD0..MD15
MAS
MUDS
MLDS
MRW
GND

U224B
74LS00
G21
H28A
74LS04
MA23
MAS

U205A
74LS07
H29
U205C
74LS07
H29
U205B
74LS07
H29
CPURESD

U228A
74LS21
E19

U227B
74LS74
E20

U222B
74LS74
F20

U204B
74LS02
H27

U206E
74LS04
F19

U191A
74LS32
G24
MAS
A20
XCS

U191B
74LS32
G24
MAS
MA23

U191D
74LS32
G24
XCS

U220C
74LS02
J21
DMAINIT
OBJCS

J27
U226D
74LS08
VOROMDTAC
FIXDTAC

SCVRAMCS
TIM1

U192A
74LS32
F21
MUWR
MULWR
MLWR

U211A
74LS08
L25
MULWR

U206B
74LS04
MREAD1
MREAD2

U206C
74LS04
F19

U188A
74LS32
K25
DMACS

U192B
74LS32
F21
MUWR

U88
27C010
F27
MA1..MA16
MD8..MD15
PGM
VPP
CE
OE
ROMCS

U139
G25
MUWR
R/W
CS2
CS1
WORKCS

6264SL

U137
27C010
E27
MD0..MD7

U105
E25
MUWR
WORKCS

6264SL

A[1..20]
D[0..15]
MA[1..14]
MD[0..15]

74LS367
U113
E26
3-StateHexBuffers

74LS245
U107
F24
buffers
A->B
CE

74LS245
U96
F23
buffers
MREAD1

74LS245
U141
E24
buffers
MREAD1

U224A
74LS00
G21
ROMCS
WORKCS

U223B
74LS139
H23
decoder
FIXCS
SCVRAMCS
OBJCS

H28F
74LS04
A20

U189A
74LS32
H24

U128
74LS138
address decoder
ROMCS
WORKCS
COLORCS
IOCS
SYSWR
VRCSET

U226A
74LS08
J27

U223A
74LS139
H23
decoder
XCS
SVCHAROMCS
OCHAROMCS

U191B
74LS32
G24

H28B
74LS04

U211C
74LS08
L25
CPUCLK

U209A
H21
74LS74

U209B
H21
74LS74
VOROMDTAC

U3
74LS138
K26
OBJSCRH
OBJSCRV
V1SCRH
V2SCRH
SYSWR
MULWR

U89
74LS273
O19
flip flops
VFLIP
HFLIP
PI0
PI1
EXRES

FIXCS
SCVRAMCS
OBJCS
0x10 0000.. 0x1F FFFF
0x12 0000.. 0x13 FFFF
0x14 0000.. 0x15 FFFF
Fixed layer RAM
Tile map 1 and 2 RAM
Sprite RAM

XCS
SVCHAROMCS
OCHAROMCS
0x00 0000..0x1F FFFF
0x40 0000..0x5F FFFF
0x60 0000..0x7F FFFF
Tile ROM
GFX ROM

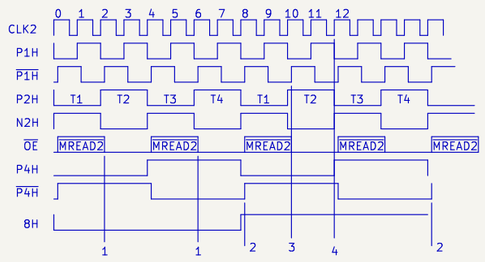
ROMCS
WORKCS
COLORCS
IOCS
SYSWR
VRCSET
0x0 0000.. 0x3 FFFF
0x4 0000.. 0x5 FFFF
0x8 0000.. 0x9 FFFF
0xA 0000.. 0xB FFFF
0xC 0000.. 0xD FFFF
0xE 0000.. 0xF FFFF
ROM
Communication RAM
Palette
System I/O
System registers
(It can also be used for read) DMACS is activated)
Graphics bank

CPU
Konami GX870
Sheet: /CPU/
File: cpu.kicad_sch
Title: The Final Round
Size: A3
Date: 2024-09-08
KiCad E.D.A. 8.0.5
Rev:
Id: 2/12

the four bits of each layer has the following meaning:
0000: Transparent
1111: Shadow
Else: Opaque

Color Mixer	
Konami GX870	
Sheet: /Color mixer/	
File: color_mixer.kicad_sch	
Title: The Final Round	
Size: A3	Date: 2024-09-08
KiCad E.D.A. 8.0.5	Id: 3/12

CLK2: 6.1397MHz
1V: 8.00kHz
2V: 4.00kHz
4V: 2.000kHz
P1H: 3.0719MHz
P2H: 1.536MHz
P4H: 0.767971MHz
8H: 384.000kHz
16H: 192.000kHz



RAM Control

Slot	Control
T1,T3	CPU
T2,T4	Video circuit

Time

1	Data output from RAM to Video circuit
2	Data latched into first DFF's
3	Data latched into second DFF's.
4	Pixel data output to FX[0..7]

The delay from the LSB (8H) of the RAM accessed by the video circuit is six pixels to output.

8-bits of data is latched every P4H cycle.
It is 2 pixels = 8bits / (4 bits / pixel).
The current pixel is multiplexed with P1H (also HFLIP).

B0 goes to pin 36 on timing circuit

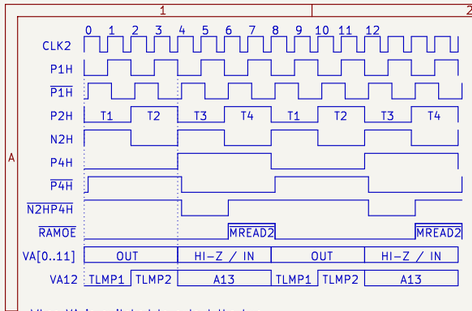
Kolla 8V och 256H

When P2H is high I1 is selected.

FXD[0..8]: Tile number
FXD[9..12]: Palette color
FXD[13]: HFLIP, not used on Final round
FXD[14]: VFLIP, not used on Final round

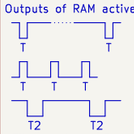
RAM
A[0..5] Horizontal position 0..511
A[6..10] Vertical position 0..255 in steps of 8
A[11.12] Tilemaps? Up to 4.

Fixed Layer
Konami GX870
Sheet: /GFX - Fixed layer/
File: GFX-Fixed layer.kicad_sch
Title: The Final Round
Size: A3 | Date: 2024-09-08 | **Rev:**
KiCad E.D.A. 8.0.5 | Id: 4/12



When VA is switched to output the bus is updated for each P1H cycle.

Outputs of RAM active when P4H is low, i.e during T1 and T2.



SYNCRLDV: Usually 60.604Hz
T is ~150ns.

VCNTEN: 8.0000kHz

HSYNC: 16.0000kHz, T2: 5.2175us

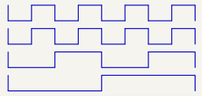
Tilemap 1 and 2
VA[5..0]: Horizontal position: 0..63 -> 0..63 x 8 = 0..504
VA[11..6]: Vertical position: 0..63 -> 0..63 x 8 = 0..504
VA12 = 0: Tilemap 1, VA12 = 1: Tilemap 2

It is unclear what SYNCRLDV (Synchronized Reload Vertical?) and VCNTEN (Vertical Counter Enable?) are used for. They become active before HSYNC goes low. They might be used to synronize the output of the vertical counters so they are updated after a write to scroll registers before HSYNC goes low, and the vertical counters.

Data latched from the RAM:

Edge	Tilemap
T1-T2	1
T2-T3	2

It is output 2 pixel clocks later to the ROMs.



The vertical counters are not the same value when they are offset with the scroll registers.

A13 = Hi: VRAM2(Tilemap 2).
A13 = Low: VRAM1(Tilemap 1)

Address	Register	
0x06	V1SCRH	Tilemap 1, horizontal scroll
0x08	V1SCRV	Tilemap 1, vertical scroll
0x10	V2SCRH	Tilemap 2, horizontal scroll
0x12	V2SCRV	Tilemap 2, vertical scroll

V1SCRV and V2SCRV are in 007779 and are accessed when SYSWR and MULWR are active.

A total of 9-bits are used for vertical scroll.
2^9 = 512 scroll positions.

V[13..15]: Color
V[11,12]: Tile bank
V[0..10]: Tile number

The 007781 is a line buffer. All counters related to all timing signals are inside the chip. It is synchronized on the rising edge of CLR and CLK2 after that.

1C[0:3]: Layer 1 Pixel color code
1C[4:6]: Layer 1 Palette
2C[0:3]: Layer 2 Pixel color code
2C[4:6]: Layer 2 Palette

Tile Layers
Konami GX870
Sheet: /GFX - Tiles/
File: GFX-Tiles.kicad_sch

Title: The Final Round

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.5		Id: 5/12

A[1..19]
D[0..15]

According to mame the object table
for each frame buffer seems to be
size 1800h = 6144 words. That is after
processing 6144 / 80 = 76 sprites

OA[0..12]

K5
U43
007783

U69 M10 6264SL
U71 M5 6264SL

L14
U75
007785

O[0..17]

U23 74LS157 L21
U18 74LS157 M19
U30 74LS157 O6
U59 74LS157 O8
U32 74LS157 O7

VPD8
VD[0..7]
HD[2..8]
HJ[0..1]
CX[0..3]
C[0..3]

U120 MX23C4100
U118 MX23C4100

Object Color Data

E[0..31]

74LS86 U193D M10
74LS04 U216E M11

74LS257 U100 P10
74LS257 U104 P11
74LS257 U102 Q11
74LS257 U98 Q10

U55 74LS374 M9
U28 74LS374 M6
U15 74LS374 M8
U53 74LS374 M7

W[0..15]
D[0..15]

OD[0..15]

U8
K9
007784

74LS245 U44 K20 buffers
74LS245 U27 K21 buffers

U189B H24 74LS32
U190C J25 74LS32

U210B 74LS74 J23

K10
U181
PAL16L8'
007793

RA9 2.2k x 8 L11
RA8 2.2k x 8 L6

SPA[0..17]

{ULRW}
{DMARW}

007783
When BEFLAG goes active it will
use the low byte OD[7..0] to create the
address for the writes to sprite RAM.
According to mame:
OA[12..0] = 0x1800 | ((OD[7..0] & 0xff) << 2)

007784
– subtracts the scroll value for the
sprites from current coordinates for the sprites.
– When DMA goes high. The 007784 will read 8 words, if bit OD15
of the first word is high then BEFLAG goes high and stays high
until the sprite RAM has processed the first four words and possibly
updated the four following words at the high end of the RAM.
it will compress the eigth words into four
words and also subtract the x– and y–scroll values from the x and y
coordinates. it will write the four words at the high end of the
sprite RAM.
– Addresses at the low end goes from 0x0000..0x0007 and the following
cycle at 0x0028..0x002F..

Scroll Register
OBJSCRV Sprite horizontal scroll
OBJSCRH Sprite vertical scroll

Data is latched in the scroll registers when
the signals OBJSCRV and OBJSCRH go low.

BEFLAG = Bus Enable FLAG?

OCHF: horizontal flip

007785
HJ[0..1] Unknown

C[0..3] Palette value for frame buffer 1
CX[0..3] Palette value for frame buffer 2

O[17:19]: RAM and ROM bank selects. Not used
on final round.

If VPD8 is high, the vertical position
is outside of visible area. The frame
buffer is not updated then.

The sprite memory is controlled in three separate cases:
1. OBJBUFINIT
2. BEFLAG
3. ~DMAINIT, when OBJBUFINIT and DMA are low.

007793

CLK21′ = CLK21

$\overline{LRW}$ = ~($\overline{DMAONS}$ & OBJBUFINIT & CLK21′ +
BEFLAG & DMA & P1H & $\overline{DMAONS}$ & CLK21′ & (~8H & HFLIP + 8H & ~HFLIP)
~$\overline{LRW}$ & ~DMAINIT)

$\overline{URW}$ = ~($\overline{DMAONS}$ & OBJBUFINIT & CLK21′ +
BEFLAG & DMA & P1H & $\overline{DMAONS}$ & CLK21′ & (8H & ~HFLIP + ~8H & HFLIP) +
~$\overline{URW}$ & ~DMAINIT)

Sprite Layer – Part 1
Konami GX870
Sheet: /Sprites1/
File: sprites1.kicad_sch

Title: The Final Round

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.5		Id: 6/12

Pixel palette data

Frame Buffer 1

Frame Buffer 2

Pixel color data

Framebuffer 1 is outputting pixel data when P256VD is low and writes to the FB is possible when P256VD is high. The conditions are reveresed for FB2.

FBWM1: usually f= 250Hz, sometimes 1.15kHz
FBWM2: Mostly active all the time. When it is not active f = 60.06Hz with small active pulses, not synced to FBWM1. It becomes non active 25 horizontal lines after VBLANK goes active.

Writes to FB are possible when the "pixel part" is sleected from the mux.
If this expression is true
(R[0:3] != 4'h0) && (FBWM1 | FBWM2 | ~WR | ~VPD8)
data is written to DRAM.

Data is written when:
– WR is active.
– Not a transparent pixel (4'h0).
– FBWM1 and FBWM2 are not active: No writes to frame buffer going on.
– VPD8 (vertical counter highest bit, VCNT > 256) is not active. Outside visible area.

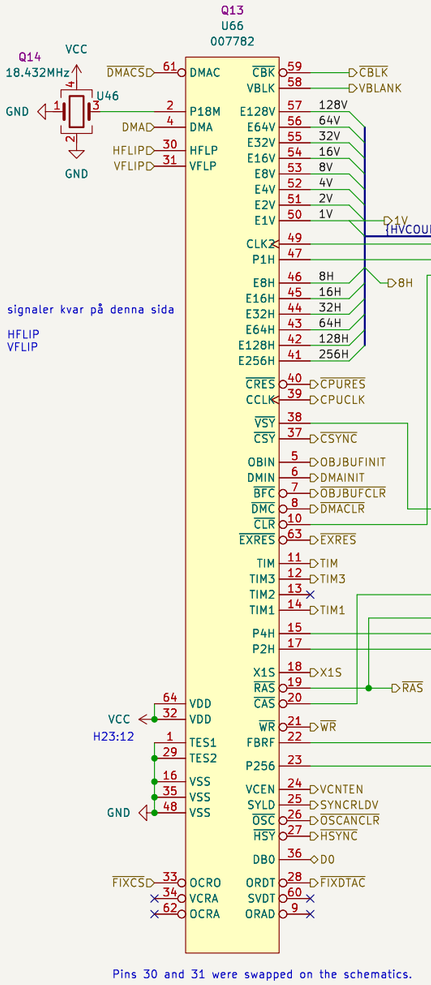
007787

Multiplexer and buffer.
– When P256VD is low R[0:27] are the active inputs. When P256VD is high r[0:27]X are the active inputs.
– It will latch 4 nibbles (R[0:3], R[8:11],..) at a time and four CLK2 cycles later output one nibble at a time at the pixel clock rate.

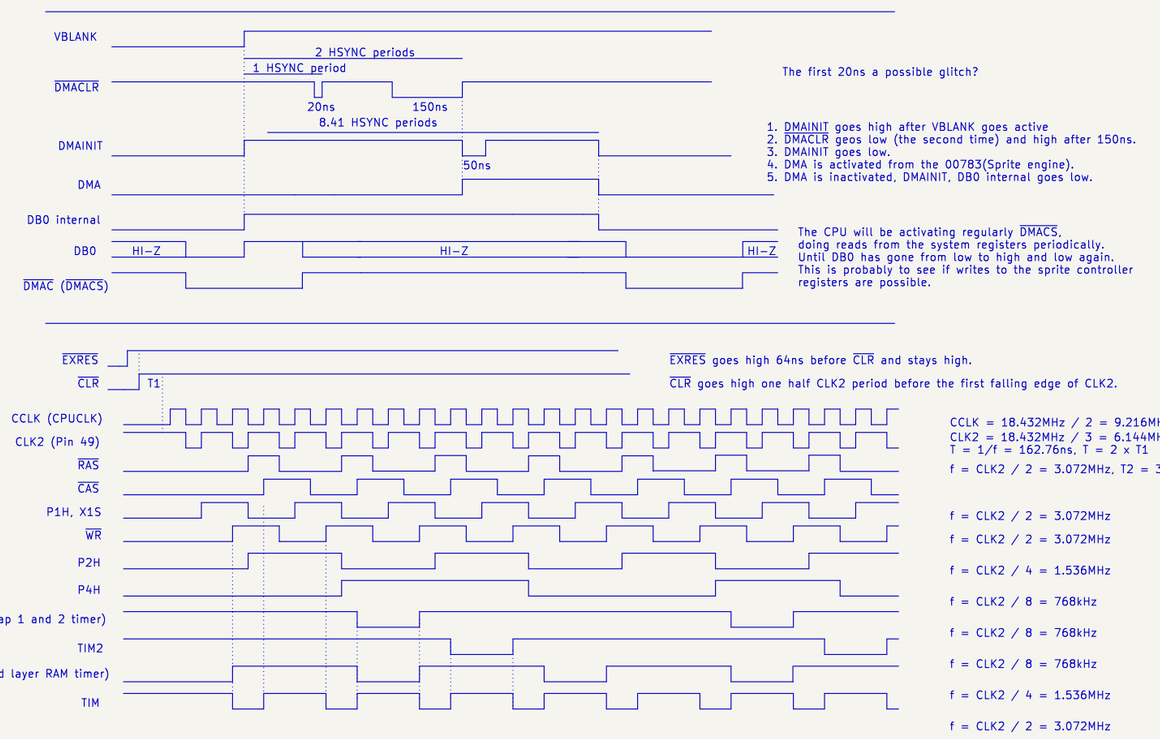
Sprite Layer – Part 2
Konami GX870
Sheet: /Sprites 2/
File: sprites2.kicad_sch

Title: The Final Round

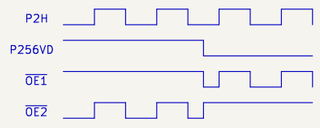
Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.5		Id: 7/12



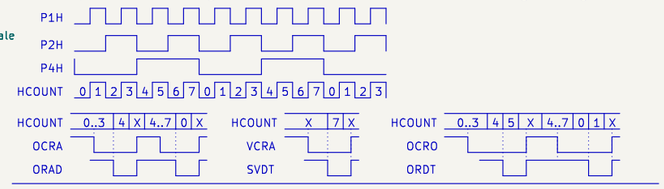
DMA Synchronization



Output enables active vs. part of frame buffer active



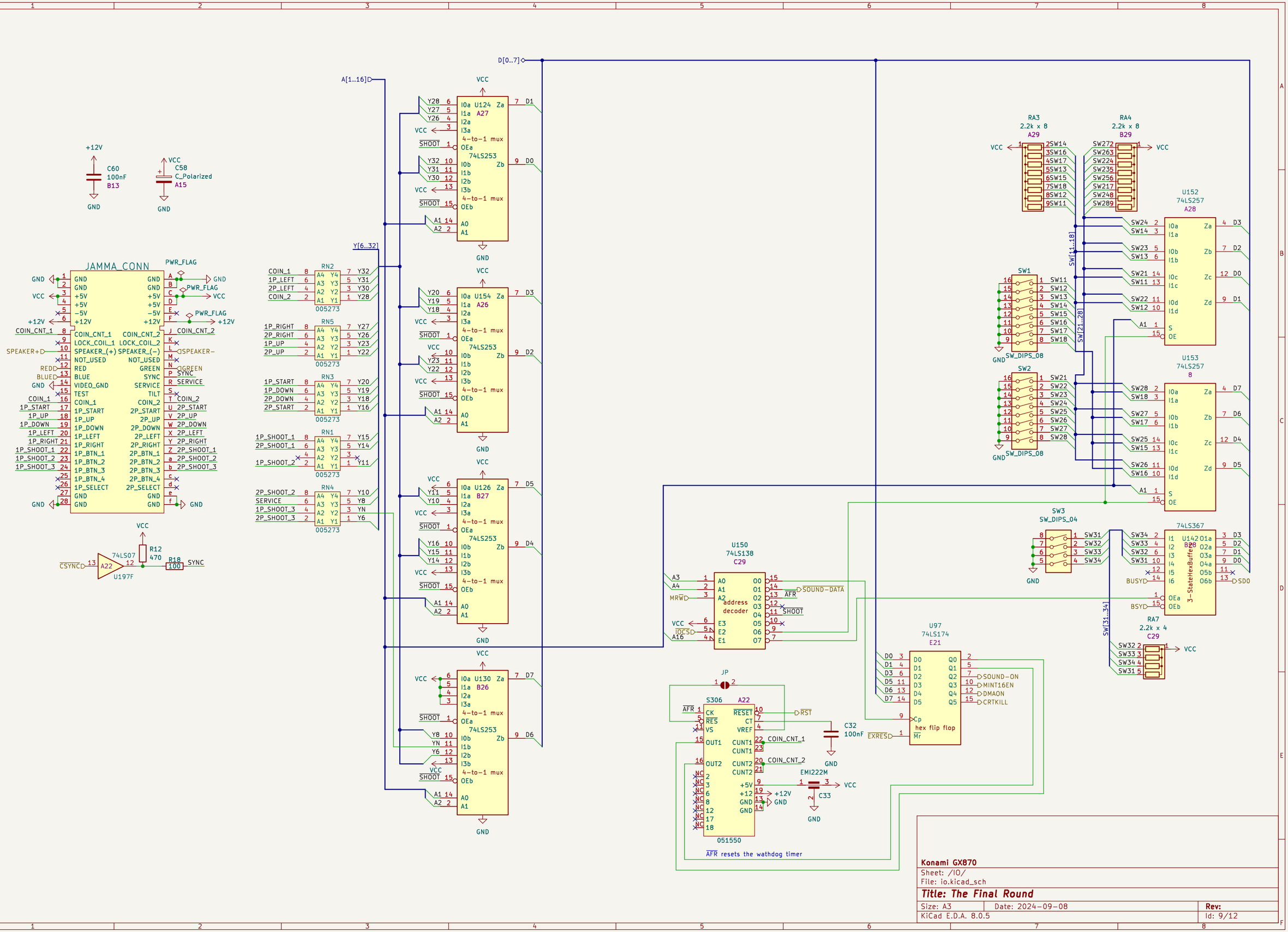
Chip select signals and Data Acknowledge signals



Timing
Konami GX870
Sheet: /Timing/
File: timing.kicad_sch

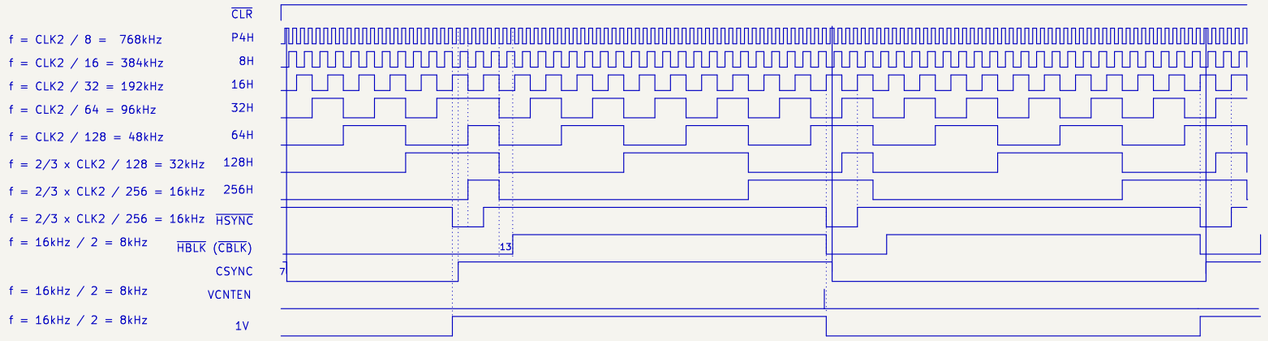
Title: The Final Round

Size: A3	Date: 2024-09-08	**Rev:**
KiCad E.D.A. 8.0.5		Id: 8/12


D[0..7]
A[1..16]
Y[6..32]
+12V
C60
100nF
B13
GND
VCC
C58
C_Polarized
A15
GND
JAMMA_CONN
PWR_FLAG
GND
VCC
+5V
-5V
+12V
COIN_CNT_1
COIN_CNT_2
LOCK_COIL_1
LOCK_COIL_2
SPEAKER_(+)
SPEAKER_(-)
NOT_USED
RED
GREEN
BLUE
SYNC
VIDEO_GND
SERVICE
TEST
TILT
COIN_1
COIN_2
1P_START
2P_START
1P_UP
2P_UP
1P_DOWN
2P_DOWN
1P_LEFT
2P_LEFT
1P_RIGHT
2P_RIGHT
1P_SHOOT_1
2P_SHOOT_1
1P_SHOOT_2
2P_SHOOT_2
1P_SHOOT_3
2P_SHOOT_3
1P_BTN_1
2P_BTN_1
1P_BTN_2
2P_BTN_2
1P_BTN_3
2P_BTN_3
1P_BTN_4
2P_BTN_4
1P_SELECT
2P_SELECT
SPEAKER+
SPEAKER-
RED
GREEN
BLUE
SYNC
SERVICE
R12
470
R18
100
74LS07
A22
U197F
CSYNC
SYNC
RN1
RN2
RN3
RN4
RN5
005273
U124 A27
U154 A26
U126 B27
U130 B26
74LS253
4-to-1 mux
U150
74LS138
C29
address decoder
SOUND-DATA
AFR
SHOOT
MRW
IOCS
A16
RA3
2.2k x 8
A29
RA4
B29
SW1
SW2
SW3
SW_DIPS_08
SW_DIPS_04
SW[11..18]
SW[21..28]
SW[31..34]
U152
74LS257
A28
U153
8
U142
B28
74LS367
3-StateHexBuffer
BUSY
BSY
SDO
RA7
C29
U97
74LS174
E21
hex flip flop
SOUND-ON
MINT16EN
DMAON
CRTKILL
EXRES
RST
JP
S306
A22
051550
AFR resets the wathdog timer
C32
100nF
EMI222M
C33
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RA6
4.7k x 4
D16
U196A
74LS04
B15
U196C
74LS04
address decoder
74LS138
U133
YM2151
U103
YM3012
2-CHANNEL SERIAL INPUT FLOATING D/A CONVERTER (DAC-A4S)
UPC324
U183D
U183C
U184A
U184B
U183A
U183B
MB3731SS
U135
A7
U151
3.58MHz
B17
U196B
74LS04
R41
1k
B18
Z80CPU
U99
C18
U227A
74LS74
E20
74LS08
U225C
D30
U127
C20
27256
U157
74LS273
B14
flip flops
U109
μPD7759
ADPCM SPEECH SYNTHESIZER LSI
ROM capability of up to 1 Mbit
640kHz
U207B
U207A
74LS74
B23
U147
74LS273
B5
571001D
U196E
74LS04
B15
U131
007232
2128SL
HM6116P-3
2kB RAM
74LS374
U145
74LS273
C15
S6
S5
007340
U159
571001D
U184C
UPC324
A11
U184D
UPC324
A11
U111
4066
U149
4066
CMOS-IC
QUAD BILATERAL SWITCH
KONAMI_007341
S4
B10
S7
B13
LA6358
U187A
U187B
U183E
U184E
U187C
RA16
47k x 4
A13
RA17
47k x 4
A10
SPEAKER+
SPEAKER-
SOUND-DATA
SDDATAEN
SOUND-ONL
SDA[0..15]
SD[0..7]
DA[0..7]
A[0..8]
SQ[0..7]
RA[0..16]
RD[0..7]
D[0..7]
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Horizontal signals

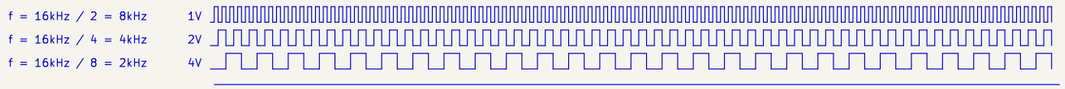


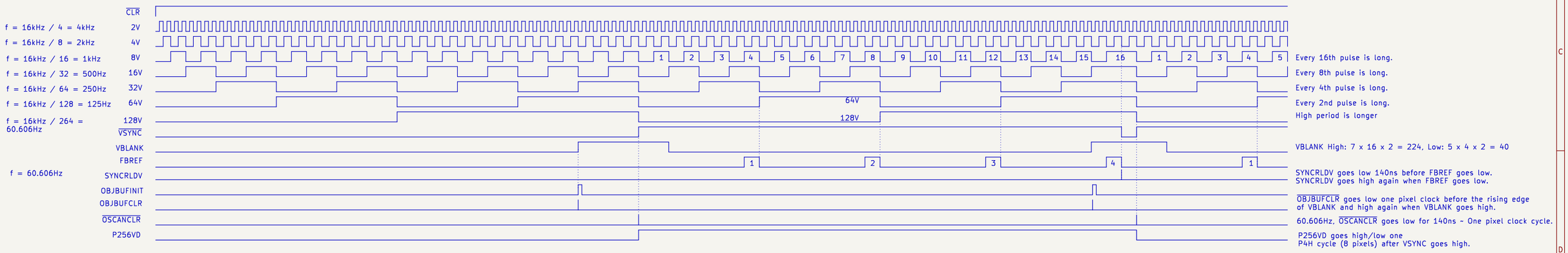
– The first VCNTEN is skipped after reset.
It goes low 140ns before HSYNC goes low,
and high again when HSYNC goes low.
VCNTEN is active right before every second falling edge of HSYNC.

– CPURES goes high, and stays high, on the seventh falling edge of HSYNC.

The first HSYNC is at HCOUNT 175 after reset.

Vertical signals

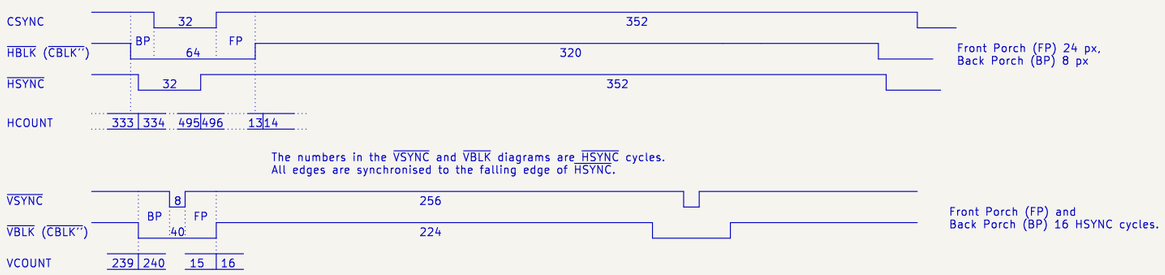




FBREF is measured at pin 22. At U186:3 FBREF is delayed by 22ns going high and 12ns going low.

Horizontal and vertical synch timing diagrams

The numbers in the HSYNC and HBLK diagrams are HSYNC cycles.
All edges are synchronised to the rising edge of CLK2.



The numbers in the VSYNC and VBLK diagrams are HSYNC cycles.
All edges are synchronised to the falling edge of HSYNC.

Pixel clock CLK2 = 18.432MHz / 3 = 6.144MHz

Horizontal lines
HSYNC freq = 6.144MHz / 384 = 16kHz
HSYNC period = 1 / 16kHz = 62.5us
Pixels / line: 384
Active pixels: 320
Blank pixels: 64

Vertical lines
VSYNC freq = 16kHz / 264 = 60.606060Hz
VSYNC period = 1 / f = 264 / 16kHz = 16.5ms
scanlines / frame: 264
Active lines: 224
Blank lines: 40

HCOUNT is bits [256H, 128H, 64H, 32H, 16H, 8H, P4H, P2H, P1H]
VCOUNT is bits [128V, 64V, 32V, 16V, 8V, 4V, 2V, 1V]

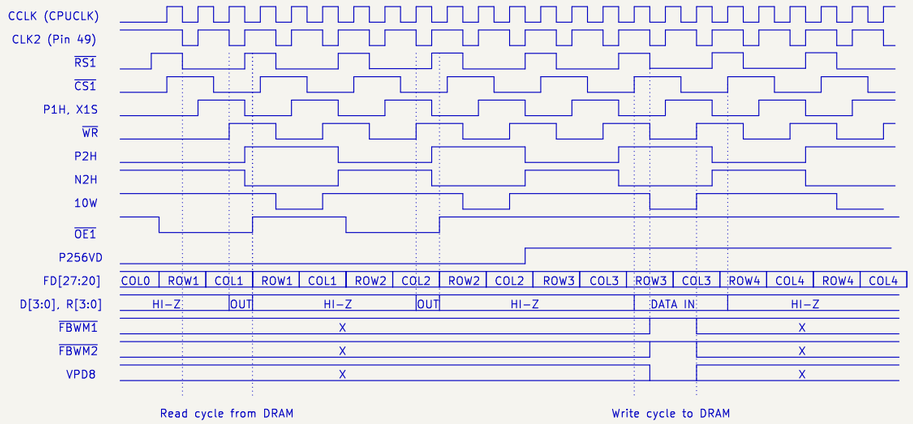
CBLK'' is at the output of color mixer.

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The same row and column addresses are repeated twice.
One for read cycle and the other for write cycles.

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